

Predicting Thyroid Cancer Risk Using Random Forest and XGBoost: Addressing Imbalanced Classes in Clinical Datasets

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Abstract—Predicting the risk of thyroid cancer early can greatly improve how well patients do. This study looks at using machine learning to predict High and Low thyroid cancer risk using data from 212,691 patients. The data includes 15 types of information like age, medical history, hormone levels, and lifestyle factors. We used two machine learning methods: Random Forest and XGBoost. Random Forest had an overall accuracy of 64.3

Index Terms—Thyroid Cancer, Machine Learning, Random Forest, XGBoost, Class Imbalance, SMOTE, Ensemble Learning, EDA

I. INTRODUCTION

Thyroid cancer is one of the most common cancers that affect the endocrine system. Finding it early and correctly determining a patient's risk level is very important for treatment. Traditional ways to diagnose include scans, hormone tests, and biopsies, which can take a long time. Machine learning provides a way to use data to predict the risk of thyroid cancer based on patient features, making risk assessment faster and more automatic.

A. Problem Statement

Accurate prediction of thyroid cancer risk levels (High, Medium, Low) is challenging due to:

- Imbalanced dataset: Medium-risk patients are underrepresented.
- Complexity of clinical, hormonal, and lifestyle features.
- Need for early and automated risk stratification to improve patient outcomes.

This research focuses on developing machine learning models (Random Forest and XGBoost) to predict High and Low-risk classes accurately and explore methods to handle the Medium-risk class imbalance. To better understand the challenges of Medium-class prediction, we perform visual analyses using 2D and 3D feature scatter plots to examine class overlap and separability.

II. DATASET AND PREPROCESSING

A. Dataset Description

The dataset, *thyroid_cancer_risk_data*, consists of 212,691 patient records with 15 features:

- **Demographic:** Age, Gender, Country, Ethnicity
- **Clinical & Lifestyle:** Family History, Radiation Exposure, Iodine Deficiency, Smoking, Obesity, Diabetes
- **Hormonal:** TSH Level, T3 Level, T4 Level
- **Other:** Nodule Size, Diagnosis (Benign/Malignant)

Categorical features were one-hot encoded (26 features). The target variable, *Thyroid_Cancer_Risk*, has three categories: High (0), Low (1), Medium (2).

B. Exploratory Data Analysis (EDA)

EDA highlighted class imbalance and feature distributions:

- Class Distribution: High-risk 6,381; Low-risk 21,678; Medium-risk 14,480.
- Continuous features (TSH, T3, T4, Nodule Size) visualized via histograms.
- Categorical features visualized using bar plots.
- Missing values replaced using median imputation.

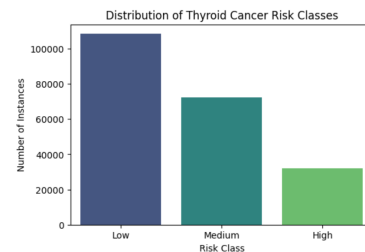


Fig. 1. Distribution of Thyroid Cancer Risk Classes (High, Medium, Low).

1) Class Distribution Visualization:

C. Train-Test Split

The dataset was split via stratified sampling:

- Training: 170,152 samples (80%)
- Testing: 42,539 samples (20%)

D. Preprocessing Pipeline

- 1) Median imputation for missing values
- 2) One-hot encoding of categorical variables
- 3) StandardScaler normalization

E. Workflow Diagram

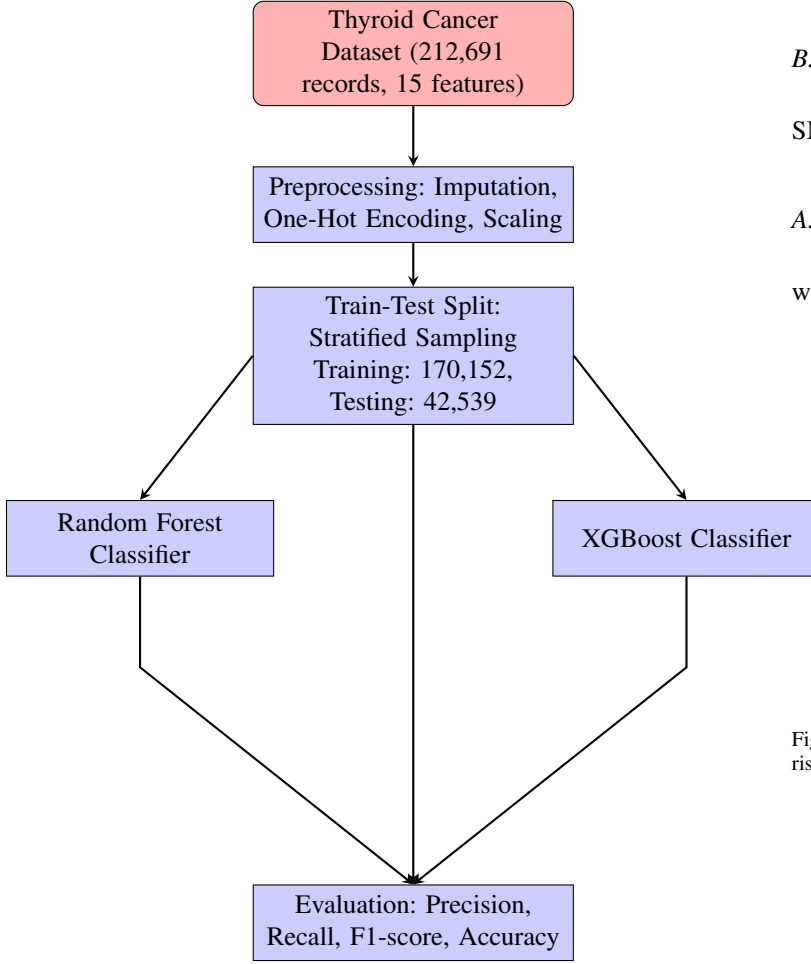


Fig. 2. Workflow of Thyroid Cancer Risk Prediction Research

III. MACHINE LEARNING MODELS USED

A. Random Forest Classifier

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of the classes. Key settings:

- Number of trees: 200
- Class weighting: balanced

B. XGBoost Classifier

XGBoost is a gradient boosting framework that builds sequential trees. Key settings:

- Number of estimators: 300
- Max depth: 4
- Learning rate: 0.05
- Objective: multi-class classification

IV. EXPERIMENTAL SETUP AND SIMULATION

A. Simulation Experiments

- 1) Train Random Forest and XGBoost on preprocessed training set

- 2) Predict on test set
- 3) Evaluate using precision, recall, F1-score, accuracy
- 4) Emphasize F1-score due to class imbalance

B. Handling Class Imbalance

Medium-risk class underrepresented; class weighting and SMOTE were insufficient.

V. RESULTS

A. 2D and 3D Class Overlap Analysis

To understand why Medium-risk class is difficult to predict, we visualize feature overlap:

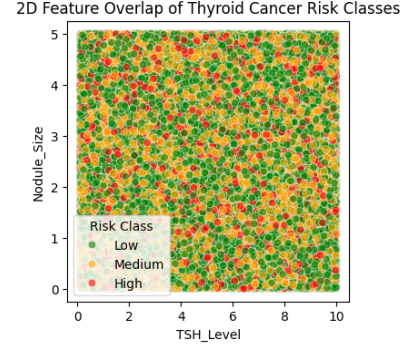


Fig. 3. 2D Scatter Plot of Feature Overlap (TSH vs Nodule Size). Medium-risk instances overlap with High and Low classes.

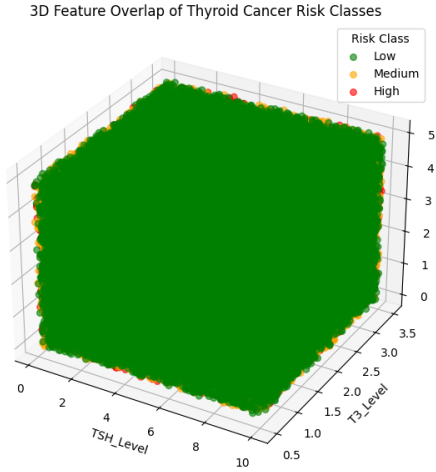


Fig. 4. 3D Scatter Plot of Feature Overlap (TSH, T3, Nodule Size). High-risk instances are separable; Medium-risk overlaps with both High and Low.

1) Visual Summary: The visual analysis clearly illustrates why the Medium-risk class is difficult to predict. High-risk instances form a distinct cluster in feature space, enabling accurate classification even with fewer samples. Medium-risk instances are scattered and overlap significantly with both High and Low-risk classes, creating fuzzy boundaries. This confirms that feature separability is a critical factor in prediction performance, often more influential than raw sample counts.

B. F1-score Performance

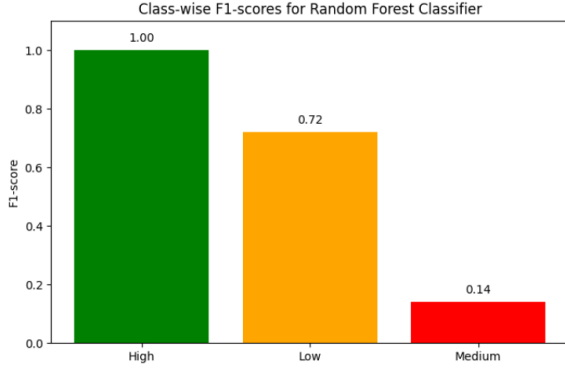


Fig. 5. Random Forest: Class-wise F1-scores.

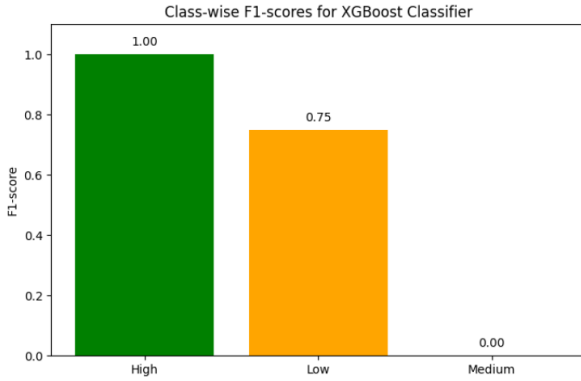


Fig. 6. XGBoost: Class-wise F1-scores.

C. Classifier Performance Tables

TABLE I
RANDOM FOREST CLASSIFIER PERFORMANCE

Class	Precision	Recall	F1-score	Support
High	1.00	1.00	1.00	6381
Low	0.60	0.91	0.72	21678
Medium	0.40	0.08	0.14	14480
Accuracy	0.64			
Macro Avg	0.67	0.67	0.62	42539
Weighted Avg	0.59	0.64	0.57	42539

TABLE II
XGBOOST CLASSIFIER PERFORMANCE

Class	Precision	Recall	F1-score	Support
High	1.00	1.00	1.00	6381
Low	0.60	1.00	0.75	21678
Medium	0.38	0.00	0.00	14480
Accuracy	0.66			
Macro Avg	0.66	0.67	0.58	42539
Weighted Avg	0.59	0.66	0.53	42539

VI. DISCUSSION AND ANALYSIS

A. Medium-risk Class Challenges

Medium-risk class has extremely low recall (0.00). Despite having more samples than High-risk, models struggle because:

- High-risk points cluster distinctly in feature space (easily separable).
- Medium-risk points overlap with both High and Low classes in 2D and 3D feature space.
- Low-risk points occupy a separate cluster.

TABLE III
CLASS COUNTS IN DATASET

Class	Count
High	6,381
Medium	14,480
Low	21,678

1) Class Counts vs Model Learning: Key insights:

- 1) High-risk is predicted well despite fewer samples → due to distinctive feature patterns (extreme hormone levels, large nodules, etc.)
- 2) Medium-risk is poorly predicted despite more samples → feature overlap causes fuzzy boundaries
- 3) Models optimize overall accuracy, often sacrificing Medium class to correctly classify High and Low

B. Limitations of the Dataset

- Severe class imbalance (Medium-risk underrepresented relative to Low)
- High feature overlap for Medium-risk
- Some features may lack discriminative power
- Potential clinical bias in historical diagnoses

C. Future Work

- Collect more Medium-risk samples
- Apply advanced oversampling or synthetic data generation
- Explore alternative models: LightGBM, CatBoost, Stacked ensembles
- Feature engineering and dimensionality reduction (PCA, t-SNE)
- Interpretability methods (SHAP, LIME) to analyze mis-classifications

VII. CONCLUSION

This study systematically processed and analyzed a thyroid cancer dataset of 212,691 patients using Random Forest and XGBoost classifiers. Steps included data preprocessing, exploratory data analysis, class distribution visualization, feature normalization, stratified train-test split, and evaluation via precision, recall, and F1-score. Ensemble models accurately classified High and Low-risk patients. Visual 2D and 3D analysis revealed that Medium-risk instances overlap significantly with both High and Low classes, explaining their poor predictive performance despite larger sample counts. Severe class

imbalance and feature overlap remain key challenges. Future research should address balanced data collection, advanced modeling, feature engineering, and interpretability to improve Medium-risk prediction.

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